UNIT 3 Work, Energy, and Power

3.1 Translational Kinetic Energy

3.2 Work

3.3 Potential Energy

3.4 Conservation of Energy

3.5 Power

3.1 Translational Kinetic Energy

Rigid body: A rigid body is an object in which the distances between any two points within it remain constant under the action of forces. In other words, the shape and size of a rigid body do not change during its motion. The motion of a rigid body can be divided into translational motion and rotational motion. The concept of a rigid body is idealized, but this idealization can provide useful approximations in many practical problems.

Translation: Translation refers to the movement of an object or coordinate system along a straight or curved path in space, where every point of the object maintains the same displacement. In other words, translational motion does not change the shape, size, or orientation of the object; it merely moves the entire object parallel to a specific direction. Translation can be linear or curved, depending on the path of motion.

Energy

- **Energy is the ability to do work.**
- **Energy is needed to make stationary objects move, to change their shape, or to warm up.**
- **When you lift an object you transfer energy from your muscles to the object.**
- **Energy is measured in Joules (J).**
- **One Joule is equal to the energy needed to raise a 1N weight through a vertical height of 1m.**

Translational kinetic energy is the energy an object possesses due to its translational motion. It can be calculated using the following formula:

$$K_E = \frac{1}{2}mv^2$$

where:

K_E **is the translational kinetic energy,**

m is the mass of the object,

v is the velocity of the object.

This formula indicates that translational kinetic energy is proportional to both the mass of the object and the square of its velocity.

3.2 Work

- **Work is done on an object when a force acting on it makes it to move.**
- **The amount of work done is dependent on the force and the distance the object moves.**
- **The work done by a force is defined as the product of the force and the distance moved in the direction of the force:**

$$W = F\Delta x$$

where x is the distance moved in the direction of the force.

Work Done by a Constant Force

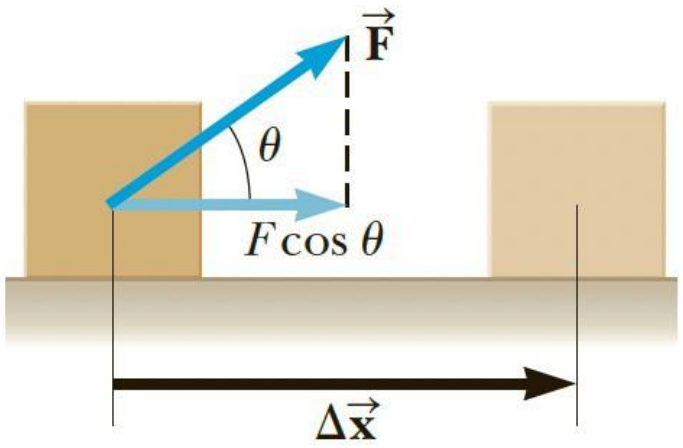
- The SI unit for work is the joule (J).
- 1 Joule is equal to the work done when a force of 1N moves its point of application by a distance of 1m in the direction of force.
- The amount of work done, calculated using

$W = F \times s$, shows the amount of energy transferred:

work done = energy transferred

1 joule = 1 newton x 1 metre

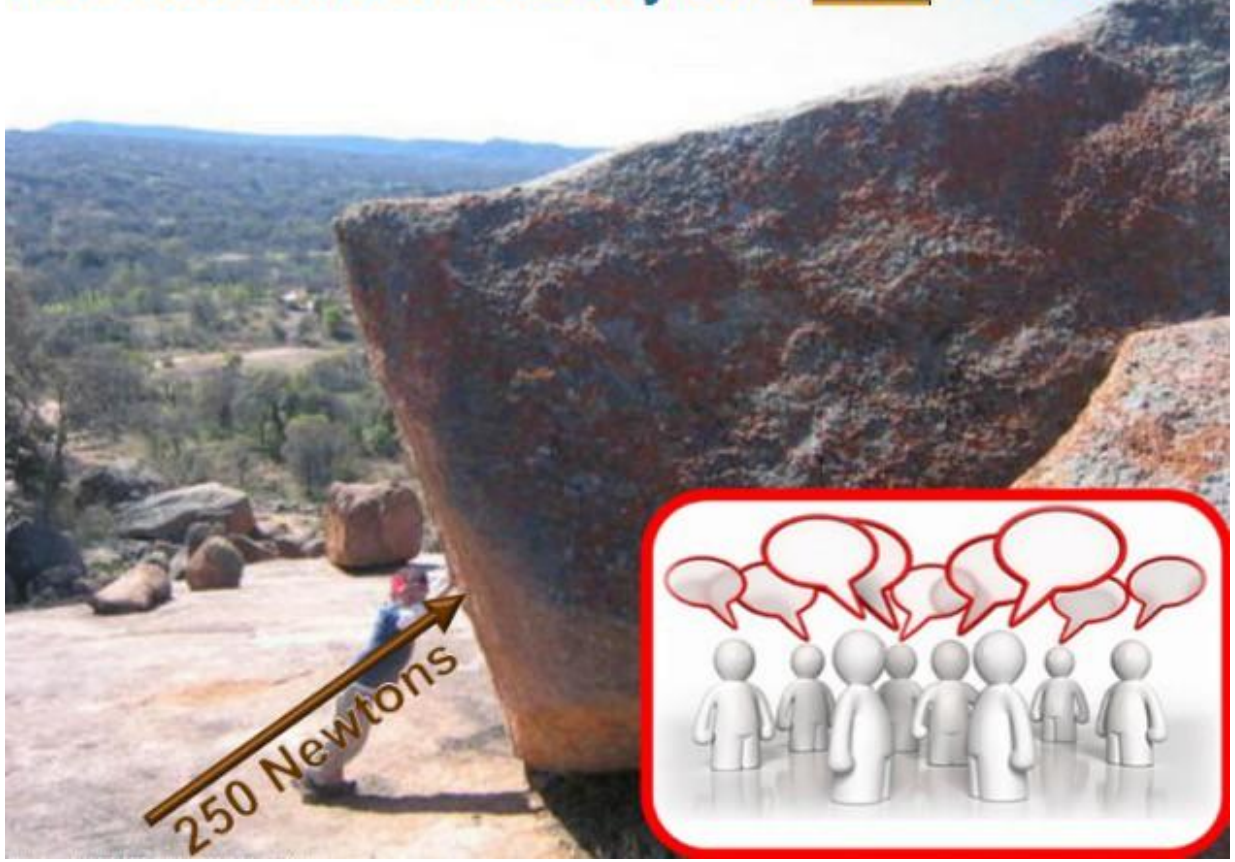
1 J = 1 N m



The magnitude of work (W) is the product of the amount of the force applied along the direction of displacement and the magnitude of the displacement.

$$W = F\cos\theta\ \Delta x$$

How much work will this boy do in three hours?



Example 7.1 Mr. Clean

A man cleaning a floor pulls a vacuum cleaner with a force of magnitude $F = 50.0$ N at an angle of 30.0° with the horizontal (Fig. 7.5). Calculate the work done by the force on the vacuum cleaner as the vacuum cleaner is displaced 3.00 m to the right.

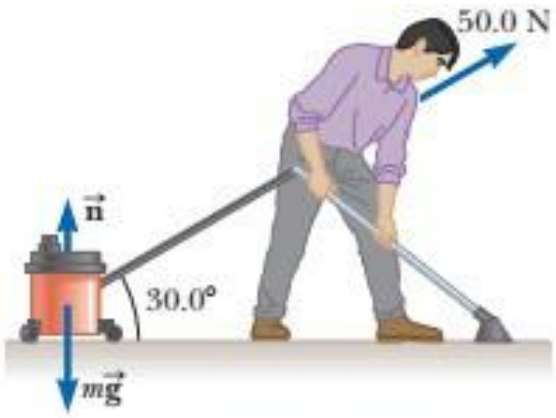


Figure 7.5 (Example 7.1) A vacuum cleaner being pulled at an angle of 30.0° from the horizontal.

Work is a Scalar (Dot) Product

Since

$$W = F\cos\theta\ \Delta x$$

and

$$A \cdot B = AB\cos\theta$$

then

$$\boxed{W = F \cdot \Delta x}$$

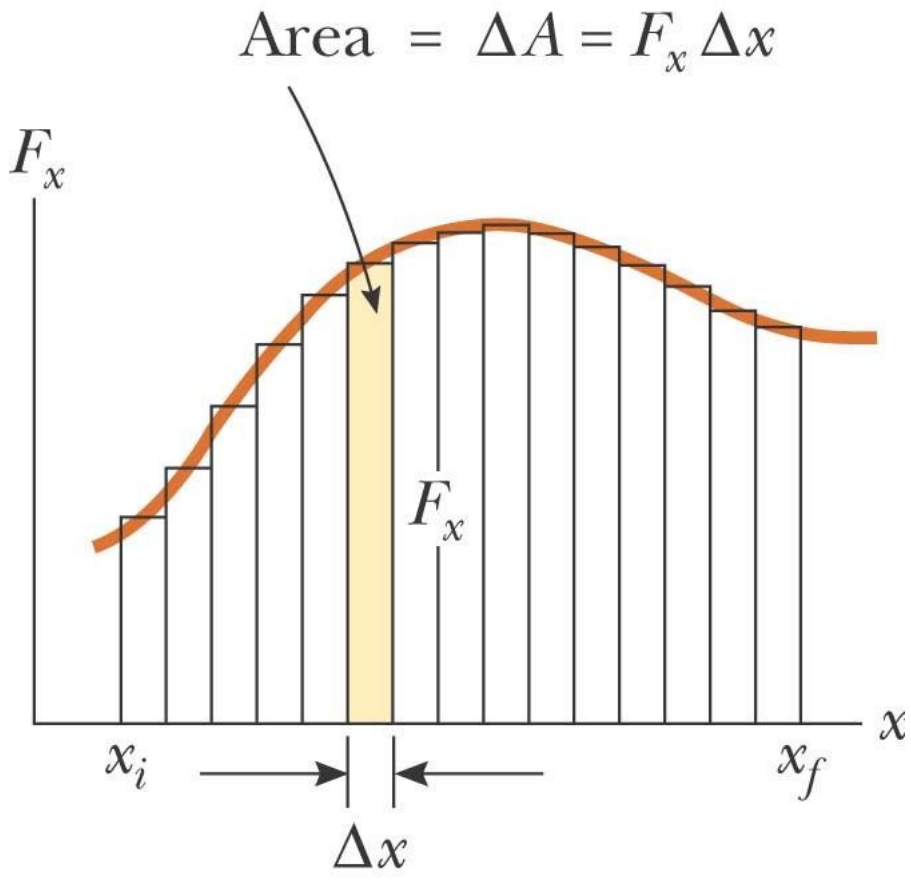
Example 7.3

A particle moving in the xy plane undergoes a displacement $\Delta x = (2.0\,\hat{i} + 3.0\,\hat{j})$ m as a constant force $F = (5.0\,\hat{i} + 2.0\,\hat{j})$ N acts on the particle.

a) Calculate the magnitudes of the displacement and the force.

b) Calculate the work done by force F.

Work Done by a Varying Force



(a)

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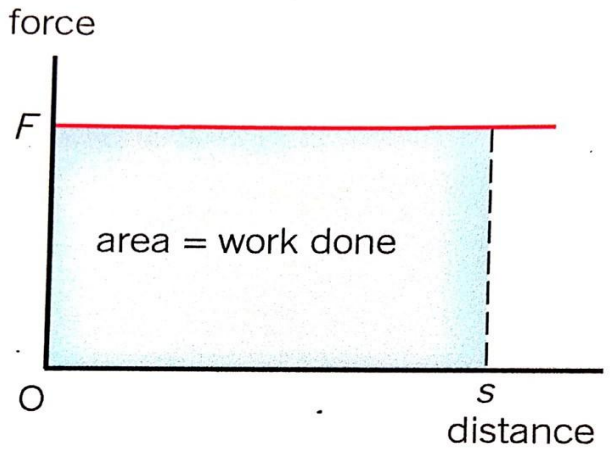
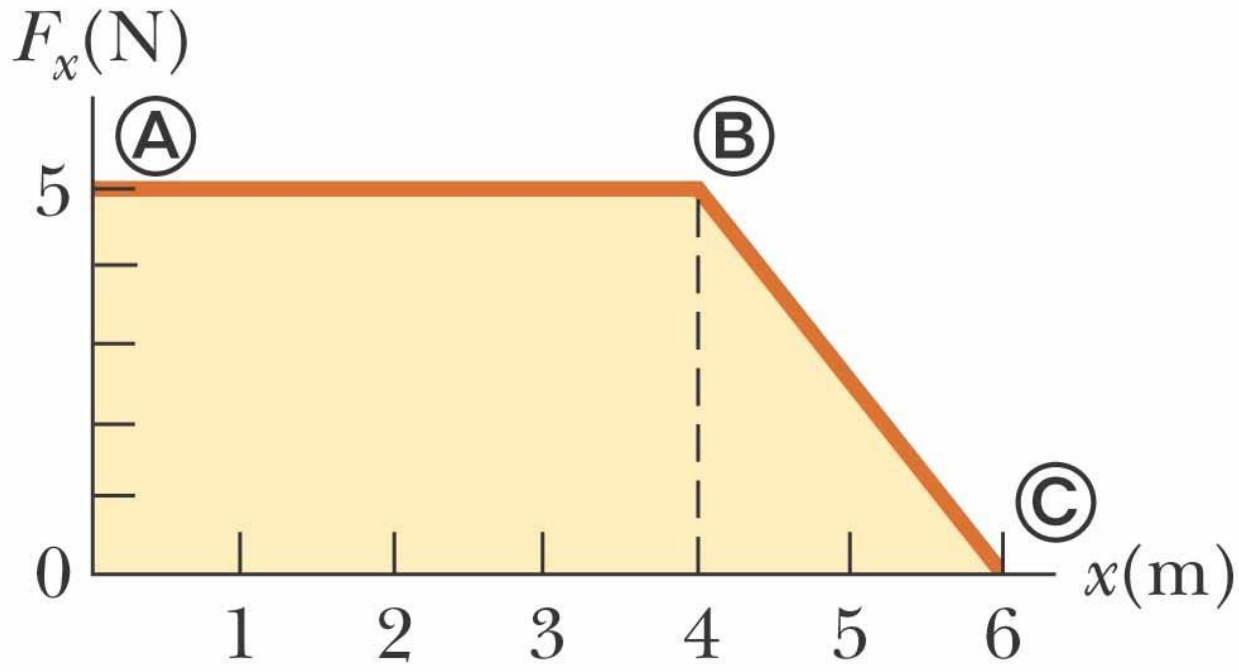
$$W = \sum_{x_i}^{x_f} F_x \Delta x$$

Force-distance graphs

- **If a constant force F acts on an object and makes it move a distance s in the direction of the force, the work done on the object, W = Fs.**
- **Figure 3 shows a graph for force against distance.**



▲ **Figure 3** *Force–distance graph for a constant force*



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A force acting on a particle varies with x, as shown above. Calculate the work done by the force as the particle moves from x = 0 to x = 6.0 m.

Kinetic Energy

$$KE = \frac{1}{2} mv^2$$

Work - Kinetic Energy Theorem

If work done on a system only changes its speed, the work done by the net force equals the change in KE of the system.

$$\Sigma W = KE_f - KE_i = \Delta KE$$

ΔKE in General

$$-f_k d = \Delta KE$$

d = length of *any* path followed

$$\Delta KE = -f_k d + \Sigma W_{\text{other forces}}$$

OR

$$KE_f = KE_i - f_k d + \Sigma W_{\text{other forces}}$$

Example 7.6 A Block Pulled on a Frictionless Surface AM

A 6.0-kg block initially at rest is pulled to the right along a frictionless, horizontal surface by a constant horizontal force of magnitude 12 N. Find the block's speed after it has moved through a horizontal distance of 3.0 m.

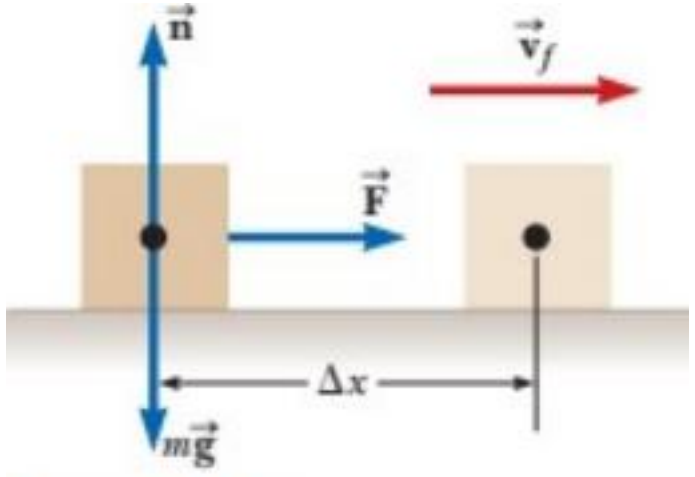


Figure 7.13 (Example 7.6) A block pulled to the right on a frictionless surface by a constant horizontal force.

3.3 Potential Energy

Conservative vs Non-Conservative Forces

There are two types of work coming from different forces: **conservative** and **nonconservative (nc)**.

•**Conservative forces** perform work that **do not depend on the path** taken (gravity, spring force). The work done by a conservative force exerted on a system only depends on the initial and final configurations of that system. The energy is converted into potential energy.

•**Nonconservative forces** perform work that **depend on the path** (friction). The energy is dissipated as heat or sound. Any non-zero nonconservative work **will change the mechanical energy**.

Potential Energy

Potential energy is the energy possessed by an object due to its position or configuration. It represents the capacity to do work as a result of its position or state. The most common types of potential energy include:

1. **Gravitational Potential Energy:**

Energy stored due to an object's position in a gravitational field.

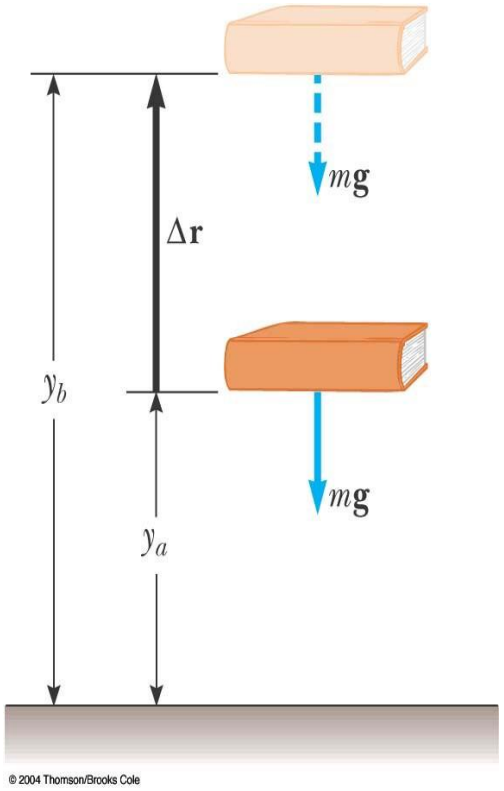
2. Elastic Potential Energy:

Energy stored in objects that can be stretched or compressed, like springs.

1. Gravitational Potential Energy:

Energy stored due to an object's position in a gravitational field.

$$U = mgh$$



where:

- **U is the gravitational potential energy,**
- **m is the mass of the object,**
- **g is the acceleration due to gravity,**
- **h is the height of the object above a reference point.**

General Form for Gravitational Potential Energy:

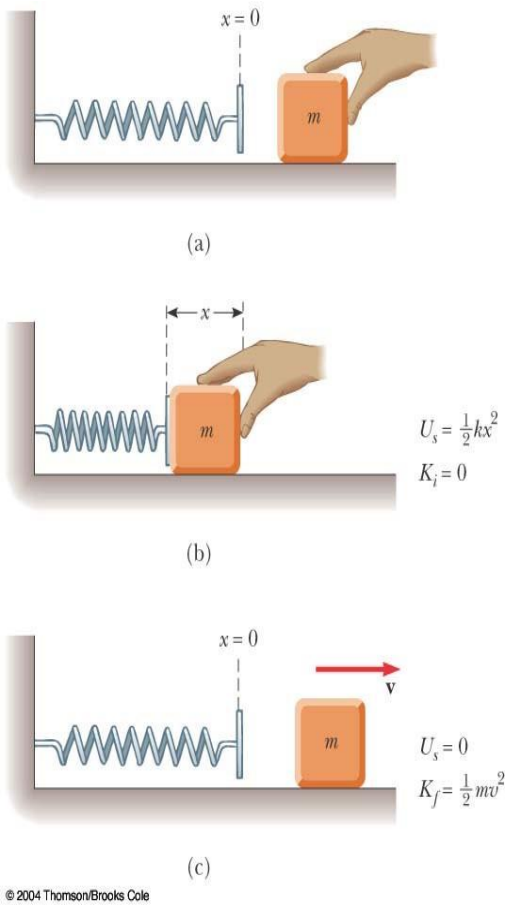
$$U_{grav} = -G\frac{m_1 m_2}{r}$$

- U_{grav} is the total gravitational potential energy. Like other energy it is measured in Joules (J).
- m_1 is the mass of the first object you are measuring U_{grav} between. The standard unit for mass is kilograms (kg).
- m_2 is the mass of the second object you are measuring U_{grav} between. The standard unit for mass is kilograms (kg).
- r is the distance between the center of mass cm of both masses m_1 and m_2. (If you don't know how to find center of mass yet, you can probably assume that its the center of whatever mass you're using). The standard unit for distance is meters (m).
- G is the gravitational constant or $6.67408 \times 10^{-11}\frac{m^3}{kg \times s^2}$

2. Elastic Potential Energy:

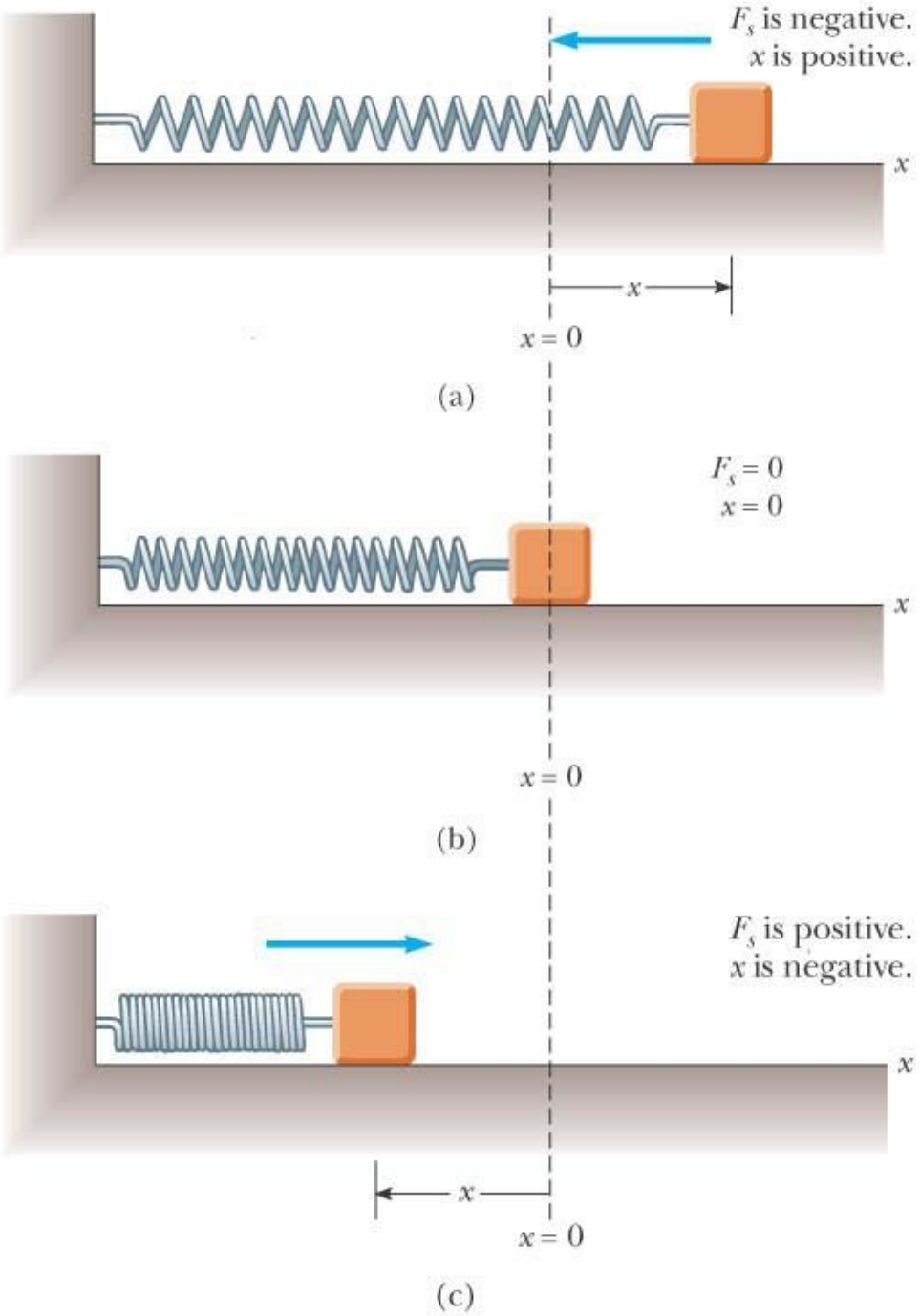
Energy stored in objects that can be stretched or compressed, like springs.

$$U = \frac{1}{2}kx^2$$



where:

- **U is the elastic potential energy,**
- **k is the spring constant,**
- **X is the displacement from the equilibrium position.**



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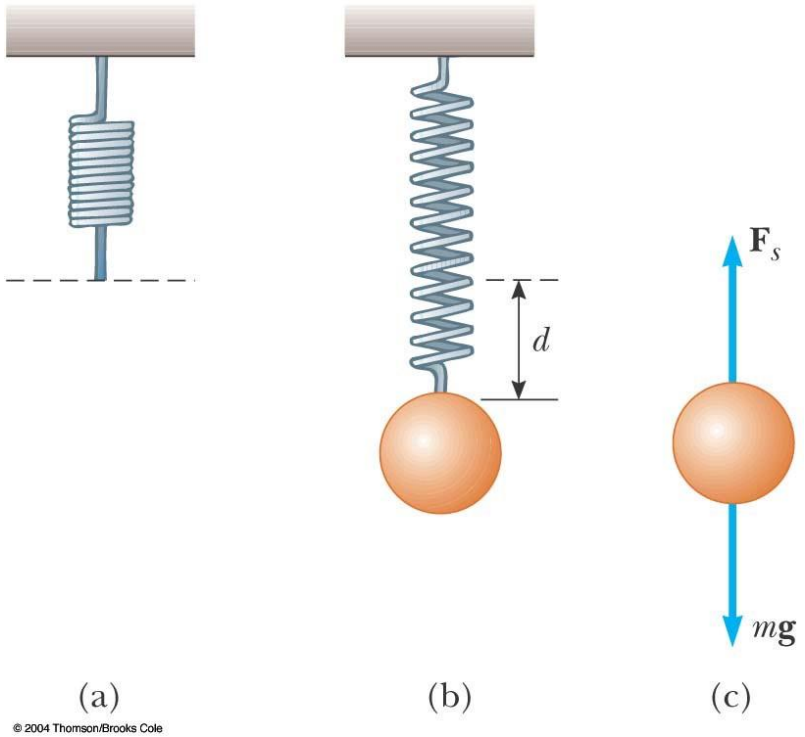
stretched spring

equilibrium position

compressed spring



A common technique used to measure the spring constant (k) is shown above. The spring is hung vertically, and an object of mass m is attached to its lower end.

Under the action of the "load" mg, the spring stretches a distance d from its equilibrium position.

a) If a spring is stretched 2.0 cm by a suspended mass of 0.55 kg, what is the spring constant of the spring ?

b) How much work is done by the spring as it stretches through this distance ?

c) Suppose the measurement is made on an elevator with an upward vertical acceleration

a. Will the unaware experimenter arrive at the same value of the spring constant ?

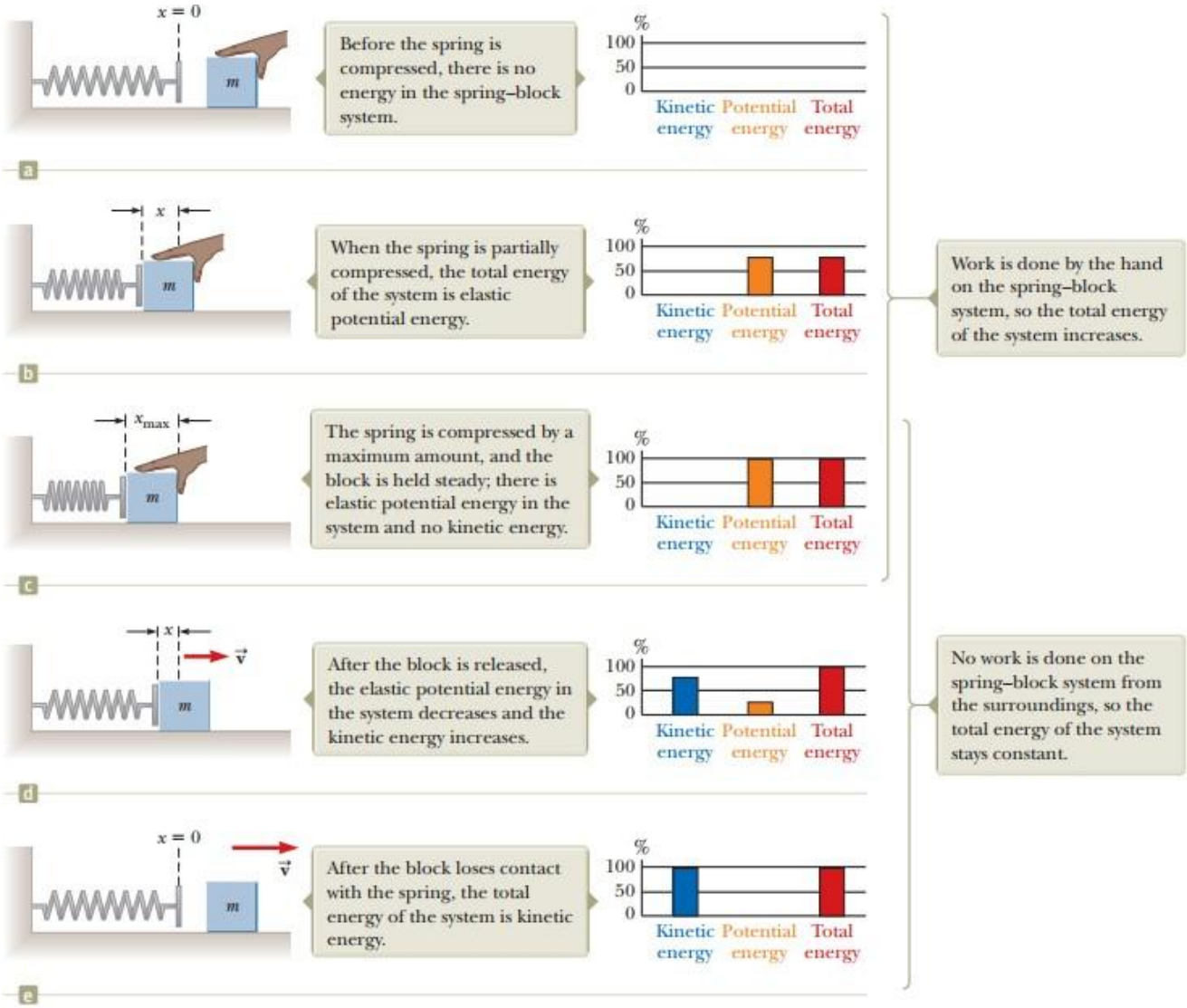
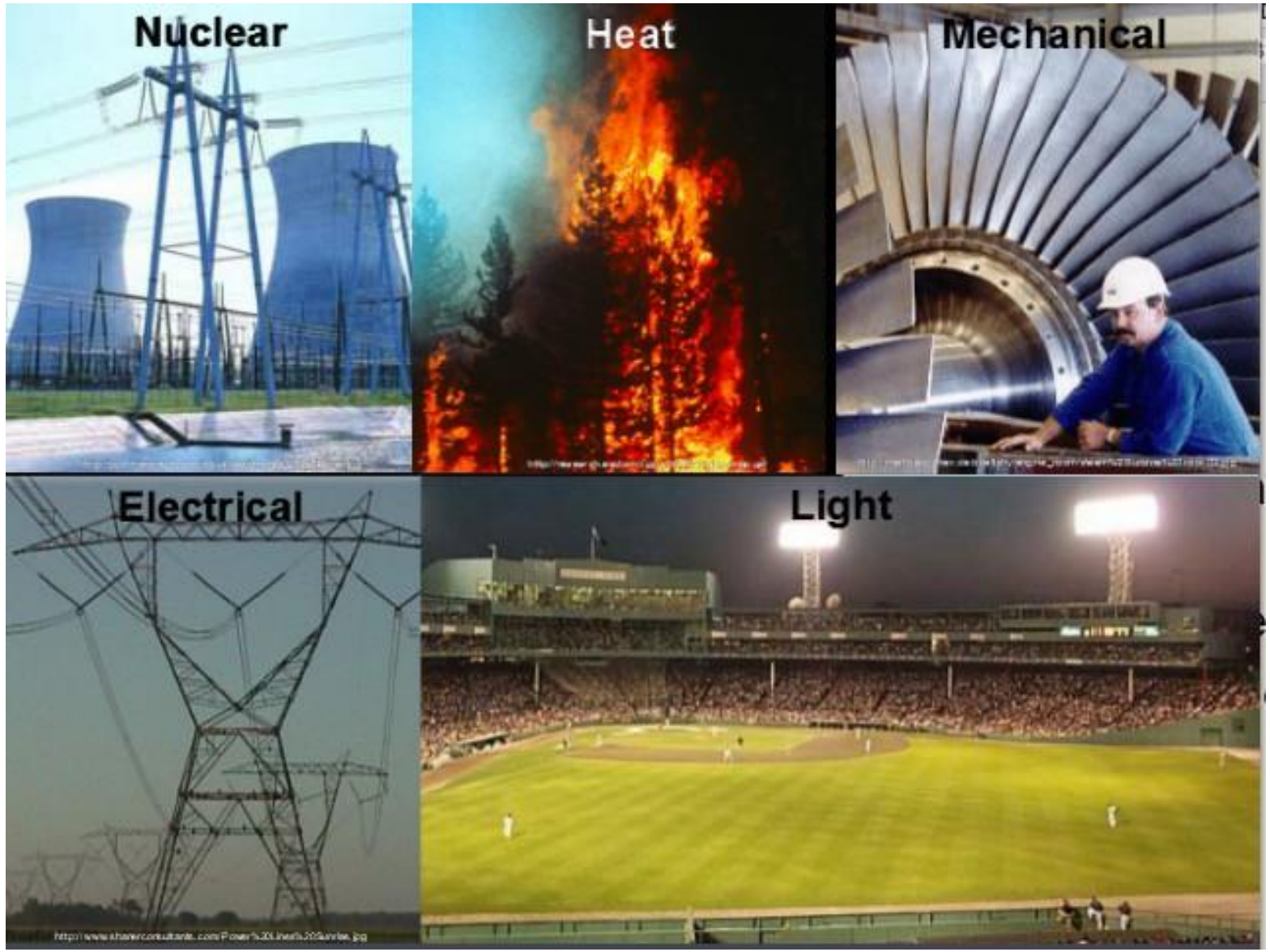


Figure 7.16 A spring on a frictionless, horizontal surface is compressed a distance x_{max} when a block of mass m is pushed against it. The block is then released and the spring pushes it to the right, where the block eventually loses contact with the spring. Parts (a) through (e) show various instants in the process. Energy bar charts on the right of each part of the figure help keep track of the energy in the system.

3.4 Conservation of Energy

- **The law of conservation of energy states that energy cannot be created or destroyed. It can only be converted from one form to another.**
- **Whenever energy is transferred, the total energy after transfer is always equal to the total amount before transfer.**
- **For example, the energy changes taking place when a battery is connected to a lamp are as follows:**

Chemical energy in battery → electrical energy in wires → light energy and internal energy of the lamp


Nuclear
Heat
Mechanical
Electrical
Light

In physics,

Mechanical Energy

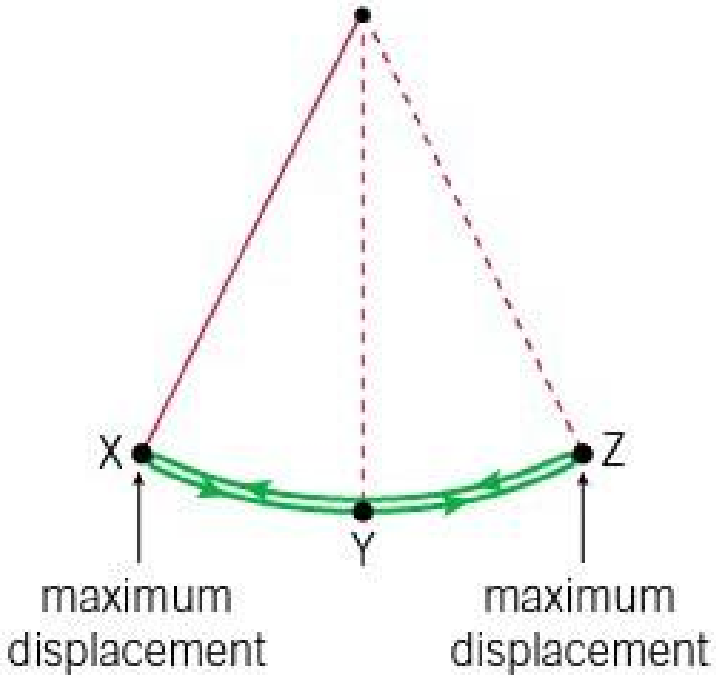
is the sum of

Potential Energy and **Kinetic Energy**.

Mechanical Energy = $mg\Delta y$ **+** $\frac{1}{2}mv^2$

It is the energy associated with the position and motion of an object.

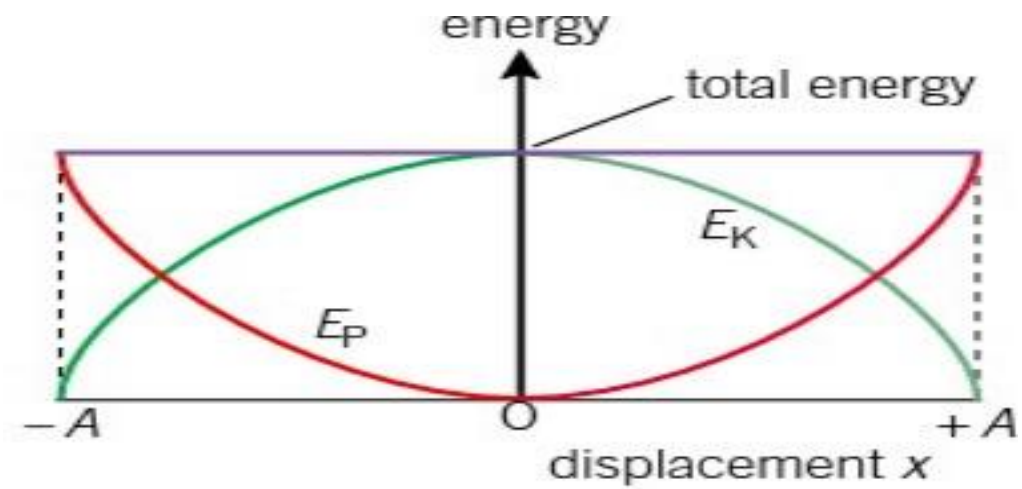
g.p.e–k.e transformations



Position	E_P	E_K
X	E_{TOTAL}	0
Y	0	E_{TOTAL}
Z	E_{TOTAL}	0

▲ **Figure 1** *The energy changes of a simple pendulum*

g.p.e–k.e transformations



▲ **Figure 2** *Energy variation with displacement*

g.p.e.–k.e. transformations

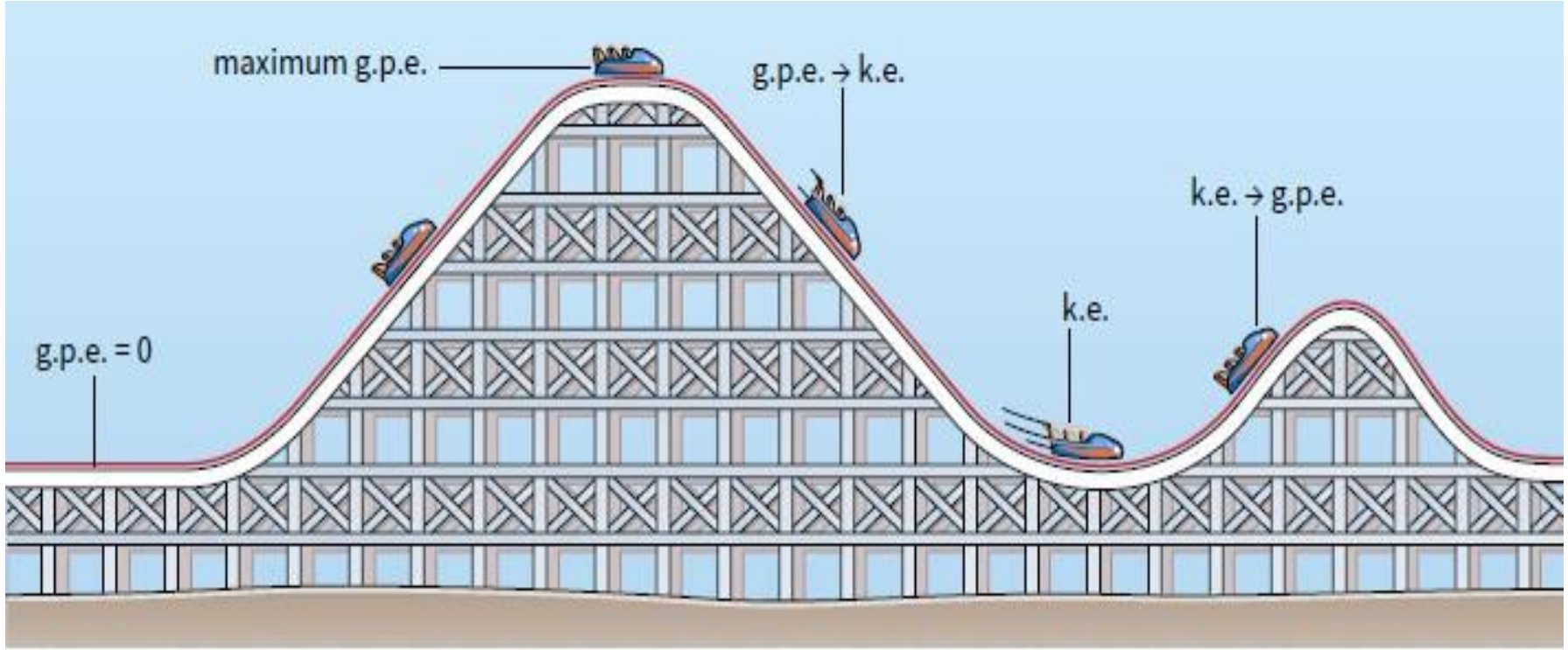


Figure 5.13 Energy changes along a roller-coaster.

$$U_{top} = K_{bottom}$$

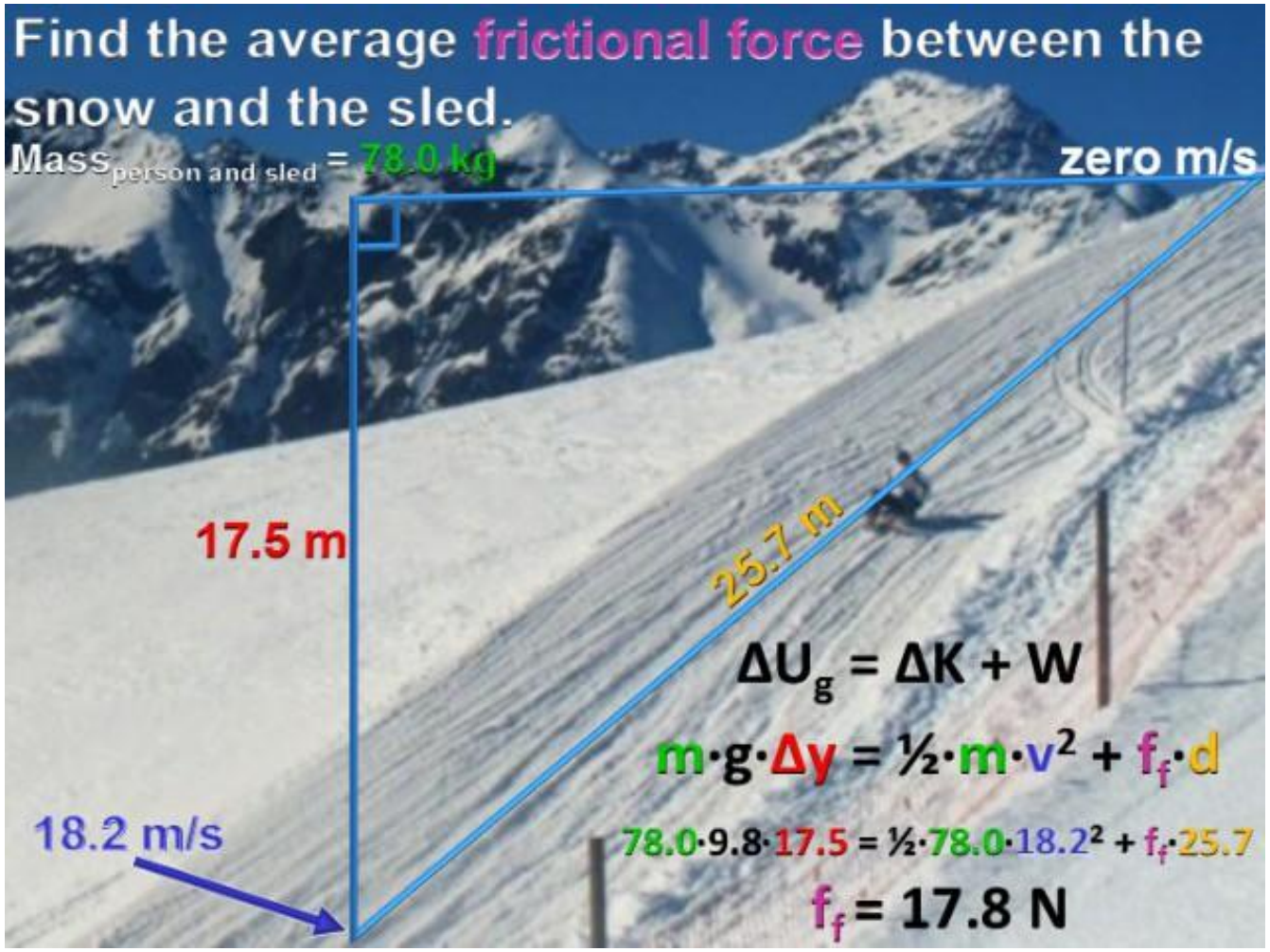
$$\cancel{m}g\,\Delta y = \frac{1}{2}\cancel{m}v^2$$

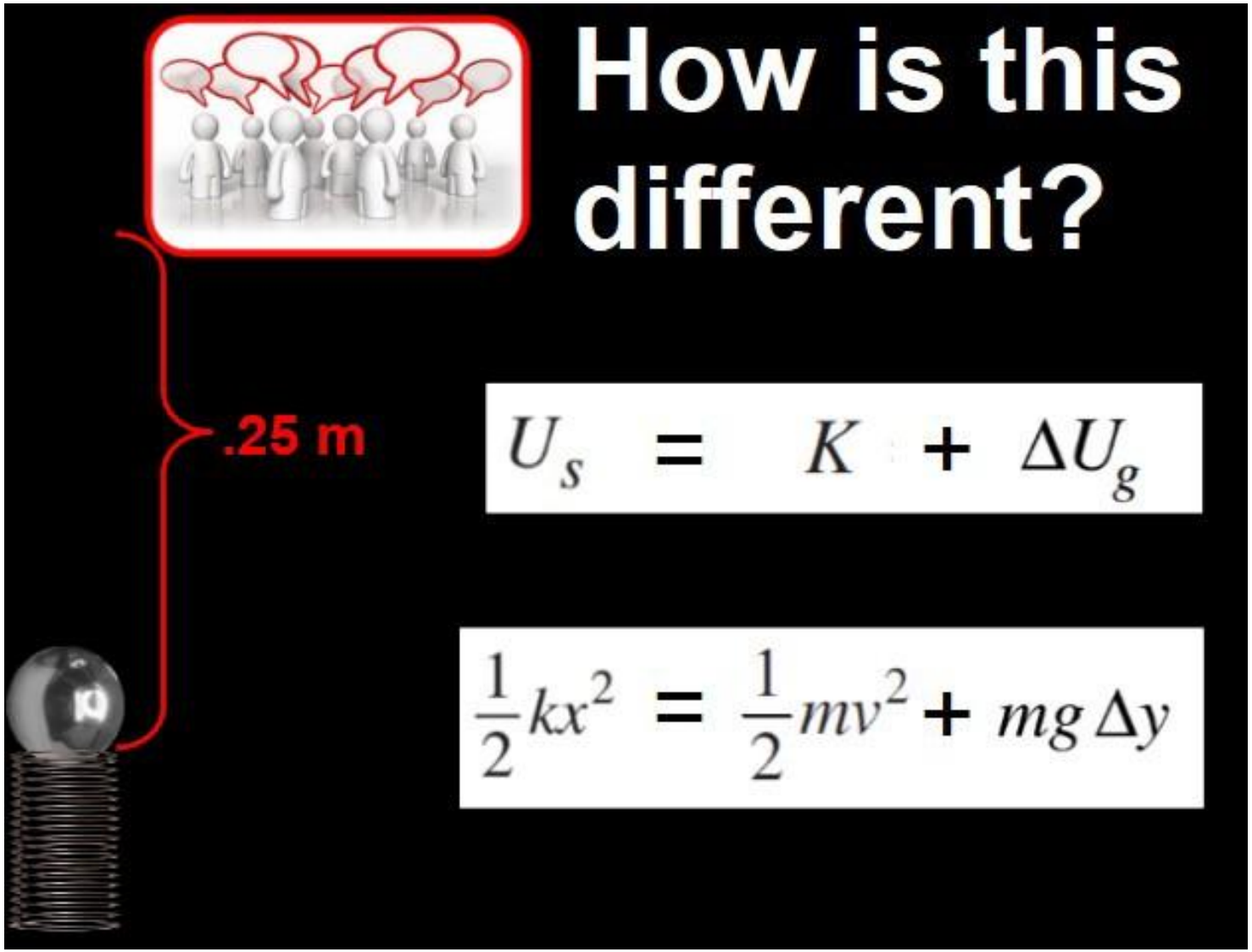
$$2 \cdot g \cdot \Delta y = v^2$$

$$v^2 = 2(9.8)(93.6)$$

$$v = 42.8 \text{ m/s}$$

$$v = 95.8 \text{ mi/hr}$$


Find the average frictional force between the snow and the sled.
Mass person and sled = 78.0 kg
zero m/s
17.5 m
25.7 m
18.2 m/s
ΔU_g = ΔK + W
m·g·Δy = ½·m·v² + f_f·d
78.0·9.8·17.5 = ½·78.0·18.2² + f_f·25.7
f_f = 17.8 N


How is this different?
.25 m
$U_s = K + \Delta U_g$
$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + mg\Delta y$

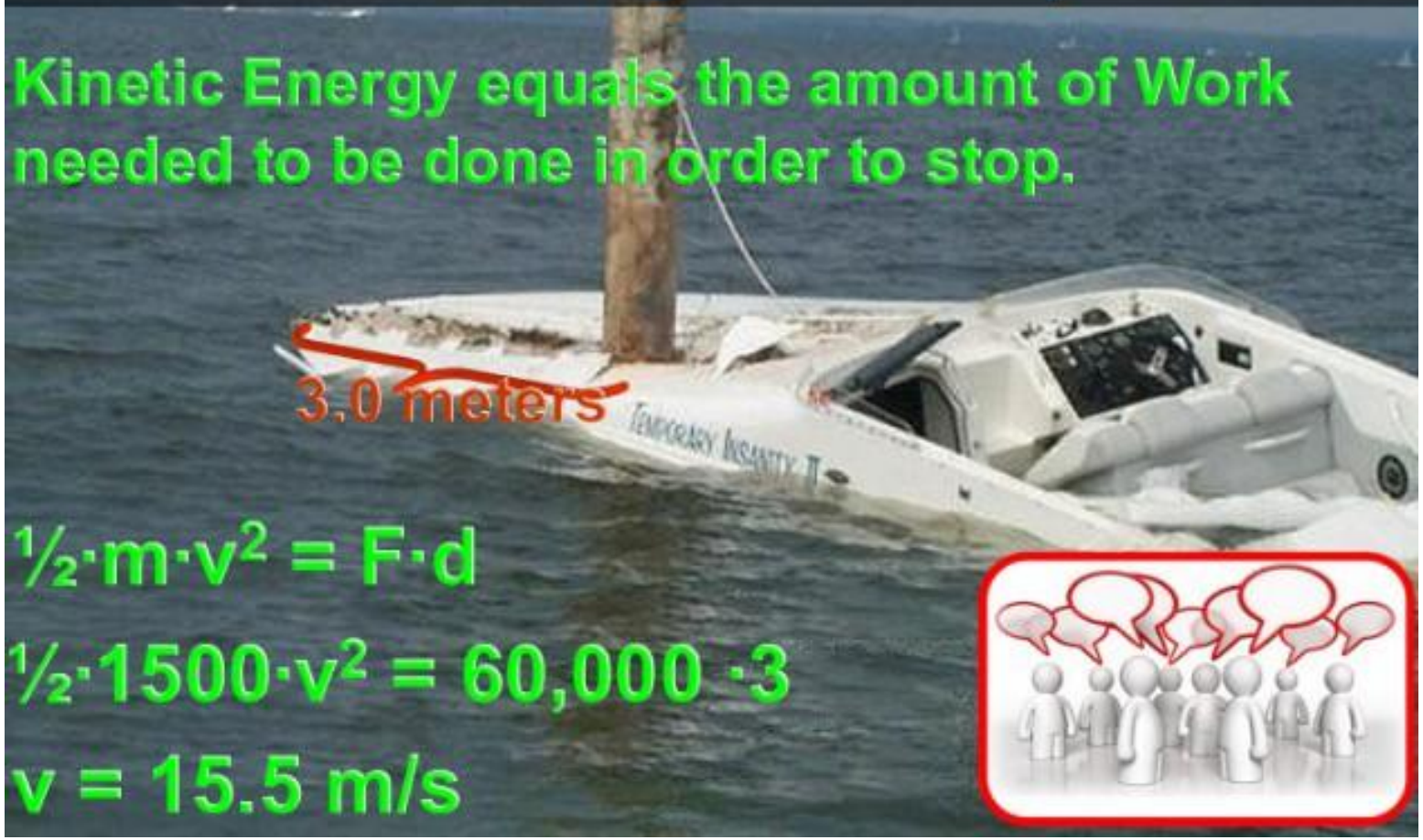
How is this different?

$$U_s = K + \Delta U_g + W$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + mg\Delta y + F_{||}d$$

If the average resistive force was 60,000 N, and the boat's mass was 1500 kg, how fast was the boat traveling before it impacted?

Kinetic Energy equals the amount of Work needed to be done in order to stop.



$\frac{1}{2} \cdot m \cdot v^2 = F \cdot d$

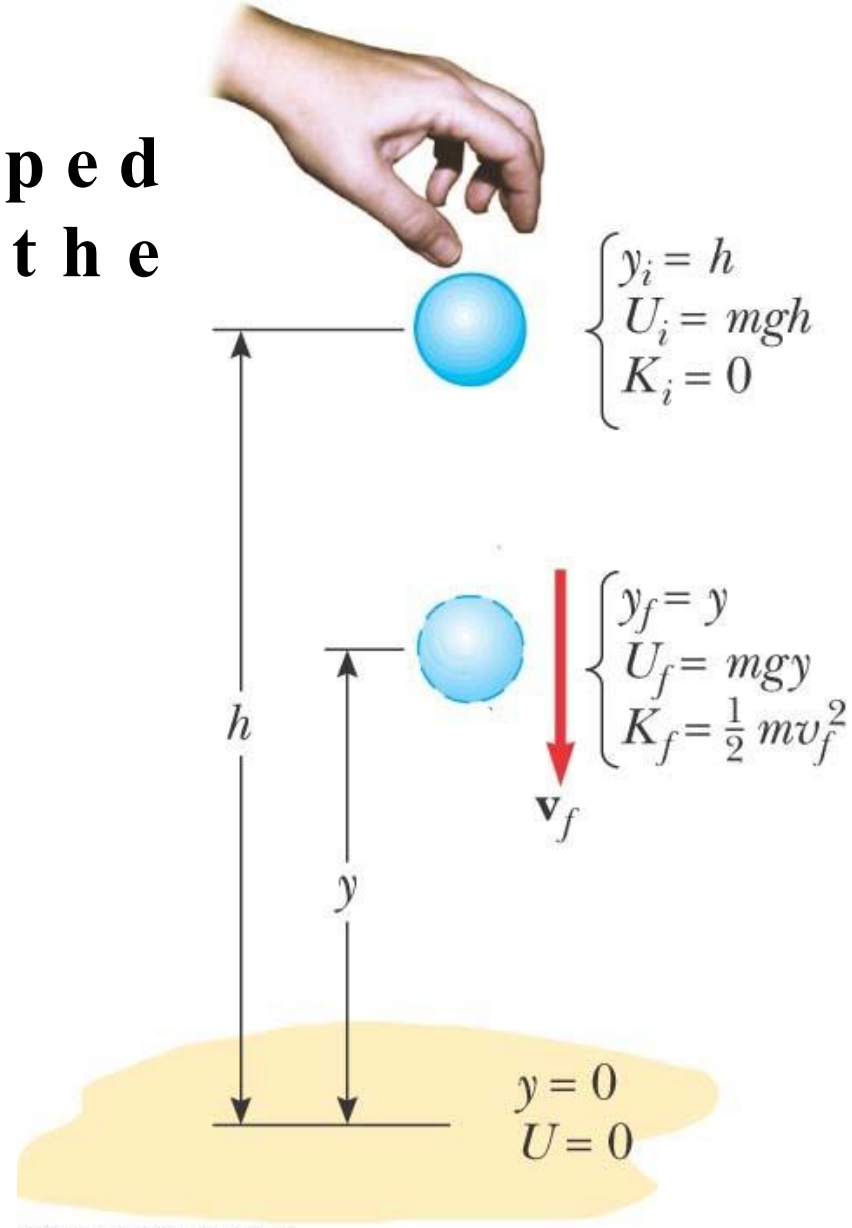
$\frac{1}{2} \cdot 1500 \cdot v^2 = 60{,}000 \cdot 3$

$v = 15.5$ m/s

Example 8.1(Ball in free fall)

A ball of mass m is dropped from a height h above the ground, as shown.

a) Neglecting air resistance, determine the speed of the ball when it is at a height y above the ground.



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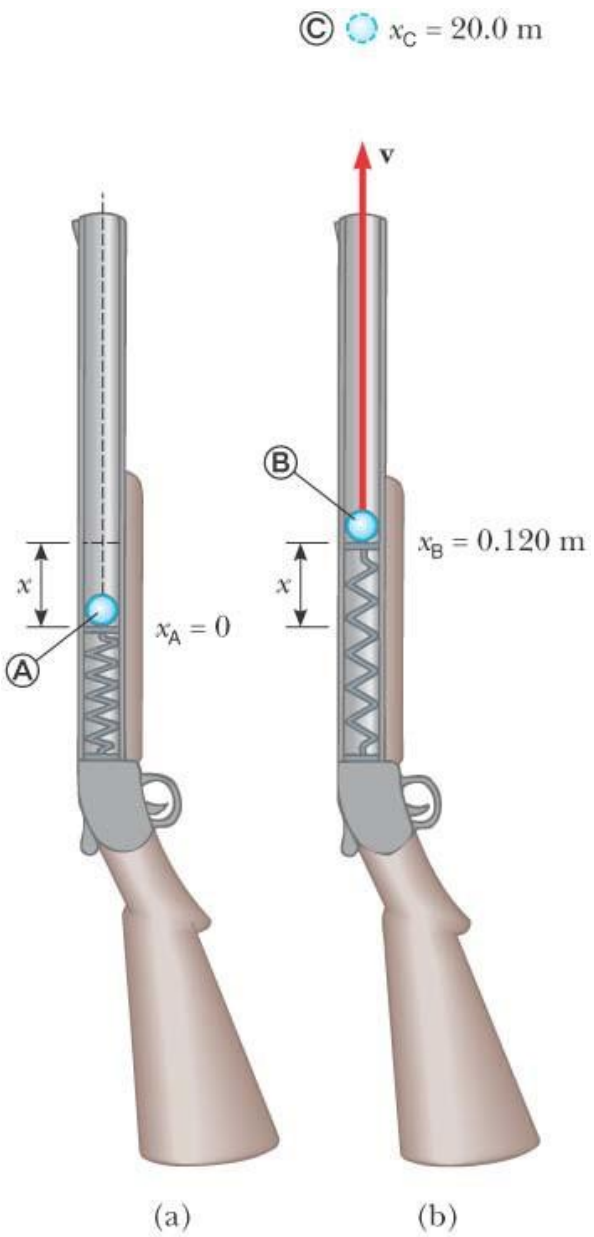
b) Determine the speed of the ball at y if at the instant of release, it already has an initial upward speed vi at the initial altitude h.

Example 8.3 (Spring loaded popgun)

The launching mechanism of a toy gun consists of a spring of unknown spring constant. When the spring is compressed 0.120 m, the gun, when fired vertically, is able to launch a 35.0 g projectile to a maximum height of 20.0 m above the position of the projectile before firing.

a) Neglecting all resistive forces, determine the spring constant.



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b) Find the speed of the projectile as it moves through the equilibrium position of the spring (where xB = 0.120 m).

Example 8.4 A Block Pulled on a Rough Surface AM

A 6.0-kg block initially at rest is pulled to the right along a horizontal surface by a constant horizontal force of 12 N.

(A) Find the speed of the block after it has moved 3.0 m if the surfaces in contact have a coefficient of kinetic friction of 0.15.

(B) Suppose the force $\vec{F}$ is applied at an angle θ as shown in Figure 8.8b. At what angle should the force be applied to achieve the largest possible speed after the block has moved 3.0 m to the right?

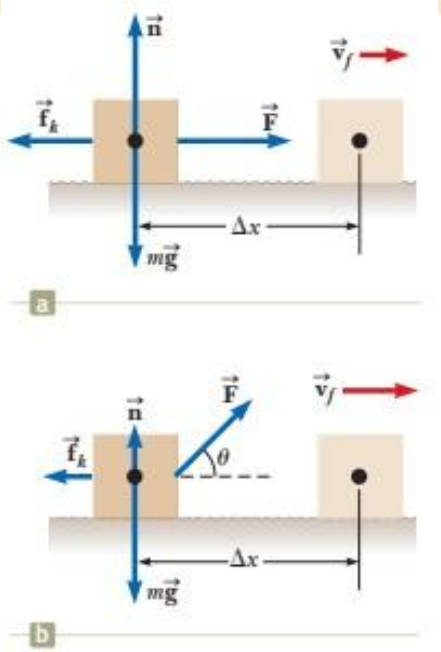


Figure 8.8 (Example 8.4) (a) A block pulled to the right on a rough surface by a constant horizontal force. (b) The applied force is at an angle θ to the horizontal.

Example 8.6 A Block–Spring System AM

A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 1 000 N/m as shown in Figure 8.9a. The spring is compressed 2.0 cm and is then released from rest as in Figure 8.9b.

(A) Calculate the speed of the block as it passes through the equilibrium position $x = 0$ if the surface is frictionless.

(B) Calculate the speed of the block as it passes through the equilibrium position if a constant friction force of 4.0 N retards its motion from the moment it is released.

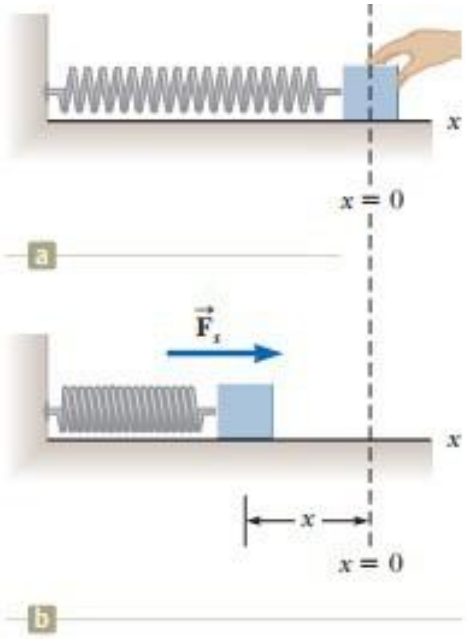


Figure 8.9 (Example 8.6) (a) A block attached to a spring is pushed inward from an initial position $x = 0$ by an external agent. (b) At position x, the block is released from rest and the spring pushes it to the right.

Example 8.7

A 3.00 kg crate slides down a ramp. The ramp is 1.00 m in length and inclined at an angle of 30.0o. The crate starts from rest at the top, experiences a constant friction force of magnitude 5.00 N, and continues to move a short distance on the horizontal floor after it leaves the ramp. Use energy methods to determine the speed of the crate at the bottom of the ramp.

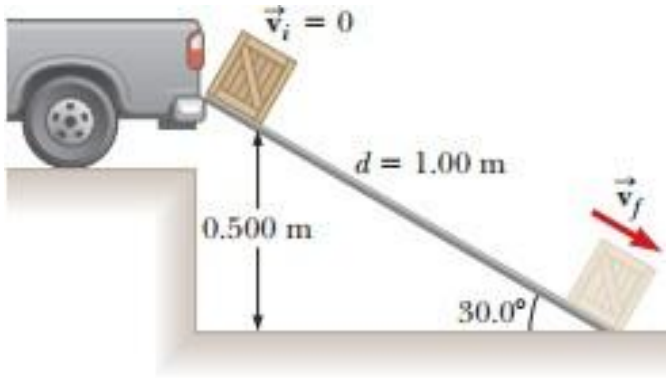


Figure 8.10 (Example 8.7) A crate slides down a ramp under the influence of gravity. The potential energy of the system decreases, whereas the kinetic energy increases.

Example 8.8 Block–Spring Collision AM

A block having a mass of 0.80 kg is given an initial velocity $v_{Ⓐ} = 1.2$ m/s to the right and collides with a spring whose mass is negligible and whose force constant is $k = 50$ N/m as shown in Figure 8.11.

(A) Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision.

(B) Suppose a constant force of kinetic friction acts between the block and the surface, with $\mu_k = 0.50$. If the speed of the block at the moment it collides with the spring is $v_{Ⓐ} = 1.2$ m/s, what is the maximum compression $x_{Ⓒ}$ in the spring?

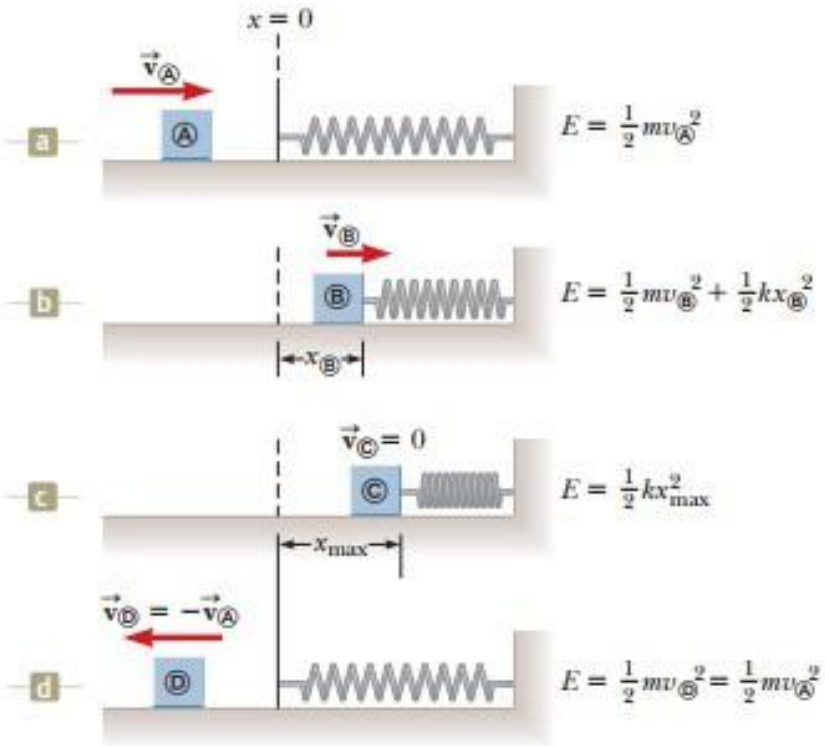
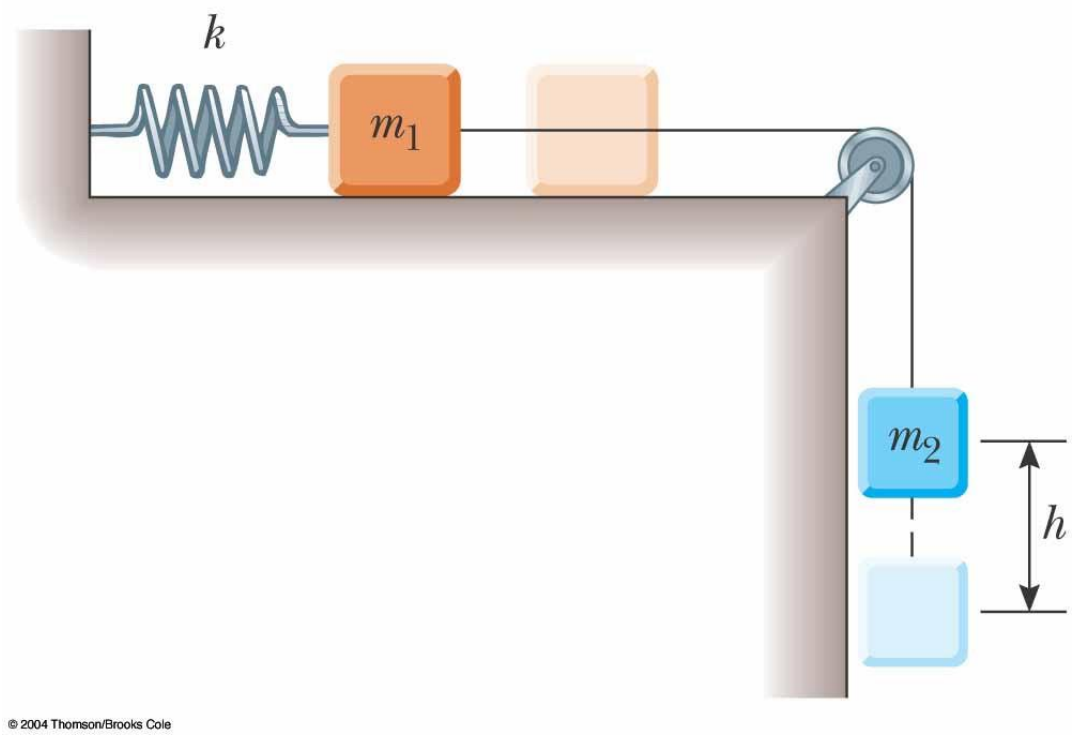


Figure 8.11 (Example 8.8) A block sliding on a frictionless, horizontal surface collides with a light spring. (a) Initially, the mechanical energy is all kinetic energy. (b) The mechanical energy is the sum of the kinetic energy of the block and the elastic potential energy in the spring. (c) The energy is entirely potential energy. (d) The energy is transformed back to the kinetic energy of the block. The total energy of the system remains constant throughout the motion.



Two blocks are connected by a light string that passes over a frictionless pulley. The block of mass m1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block m1 and the surface.

3.5 Power

- **Power is defined as the rate of energy transfer (rate of work done).**
- **The unit of power is the watt (W), equal to an energy transfer rate of 1 joule per second. Note that 1 kilowatt (kW) = 1000W, and 1 megawatt (MW) = 1000000W.**
- **If energy is transferred steadily (work is done) in time Δt,**

$$power\ P = \frac{\Delta E}{\Delta t} = \frac{\Delta W}{\Delta t}$$

Average Power

time rate of energy transfer

$$P = \frac{W}{\Delta t}$$

Instantaneous Power

$$P = \lim_{\Delta t \to 0} \frac{W}{\Delta t} = \frac{dW}{dt}$$

$$P = \frac{dW}{dt} = \frac{F \cdot dx}{dt} = F \cdot v$$

Units of Power

SI unit of power is J/s or the Watt (W).

$$1\ \mathrm{W} = 1\ \mathrm{J/s} = 1\ \mathrm{kg \cdot m^2/s^3}$$

In the U.S. customary system, the unit of power is the horsepower (hp).

$$1\ \mathrm{hp} = 746\ \mathrm{W}$$

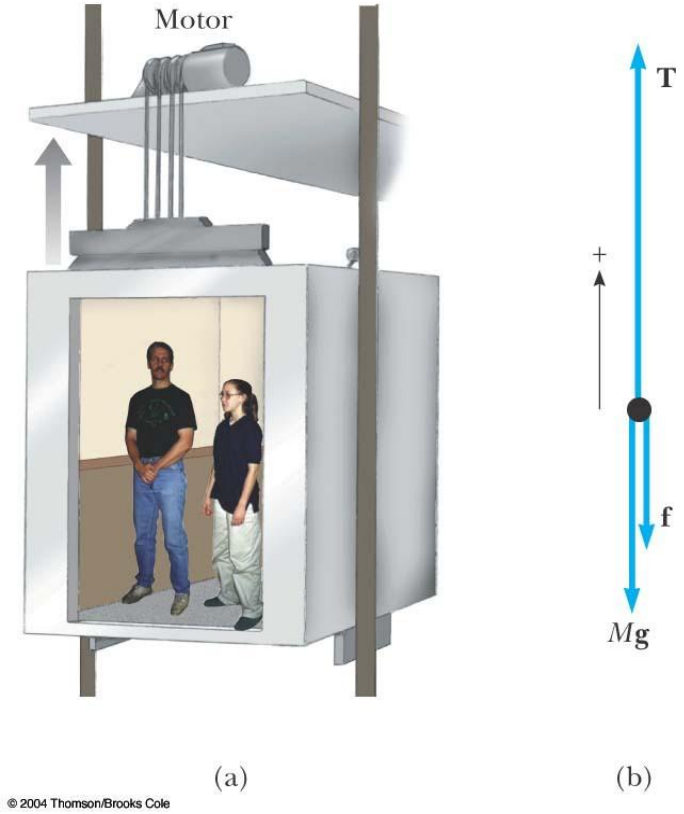
The kilowatt-hour (kWh)

The energy transferred in 1 h at the constant rate of 1kW = 1000 J/s.

$$1\ \text{kWh} = (10^3\ \text{W})(3600\ \text{s}) = 3.60 \times 10^6\ \text{J}$$

* **Note that a kWh is a unit of energy, not power.**

Example 1



An elevator car has a mass of 1600 kg and is carrying passengers having a combined mass of 200 kg. A constant friction force of 4000 N retards its motion upward, as shown above.

a) What power delivered by the motor is required to lift the elevator car at a constant speed of 3.00 m/s ?

b) What power must the motor deliver at the instant the speed of the elevator is v if the motor is designed to provide the elevator car with an upward acceleration of 1.00 m/s2 ?

Example Question of AP Classroom

A ball of mass m and momentum p has kinetic energy equal to which of the following?

(A) $\frac{1}{2}\frac{p^2}{m}$

(B) $\frac{p^2}{m}$

(C) $2\frac{p^2}{m}$

(D) $\frac{1}{2}\frac{m}{p^2}$

(E) $2\frac{m}{p^2}$

Example Question of AP Classroom

A student is asked to determine the work done on a block of wood when the block is pulled horizontally using an attached string. The student is supplied with a spring scale, a stopwatch, and a meterstick. Which of the following graphical analysis techniques will allow the student to determine the work done on the block by the string?

(A) Graphing the force as a function of time and calculating the slope

(B) Graphing the force as a function of time and calculating the area under the curve

(C) Graphing the force as a function of distance and calculating the slope

(D) Graphing the force as a function of distance and calculating the area under the curve

Example Question of AP Classroom

A nonrotating spherical planet with no atmosphere has mass M and radius R. A projectile of mass m is launched radially from the surface of the planet with initial speed $v = \sqrt{GM/2R}$. The potential energy of the projectile-planet system, as a function of the projectile's distance r from the center of the planet, is given by $U = -GMm/r$. The greatest distance from the center of the planet that the projectile reaches is

(A) infinity

(B) R

(C) $\frac{7}{5}R$

(D) $\frac{4}{3}R$

(E) $\sqrt{2}R$